

# CONTINUATION — Helix

## Proxy: VPN-Aware Dynamic Routing Extension

**Revision: 11 Last modified: 2026-07-01T18:15:00Z Status:** Active — **§11.4.126 autonomous loop** (operator: "keep hardening, don't tag yet" + "3-4 parallel subagents, rock-solid evidence, no bluff"). **Rev-11 wave (HEAD 4e2810c, all FF to origin):** the VPN-LAN service-access feature is **COMPLETE** — **all Phases 0-12 committed** incl. the operator's **Phase 12 bidirectional exposure** (2ed0fed bidirectional\_exposure.md + ingress\_allowlist\_teeth.sh, wired into the standing suite) and **Phase 1 SSRF carve-out teeth** (suite-wired). **All 5 autonomous VPN-LAN teeth GREEN** (SSRF carve-out + SSRF\_MUT, ingress-allowlist + INGRESS\_MUT, bridge). Landed also: DNS-rebinding SSRF gap demo + smokescreen design (4e2810c, 7 sources §11.4.150), control-plane coverage **acl-helper 0→53.8%** (4a9b75f) + **cmd/api 0→61.1%** (4e2810c, -race clean, go.sum untouched), §11.4.169 **stress+chaos** (97d9733, 100-iter identical hash) + **benchmark+memory** (4e2810c), deep **OSS survey** (958d110, 28 projects/50 URLs). **§11.4.1 anti-bluff win (7c4345d+4a007a0+30ec5b5):** the standing suite was HANGING forever on the LE phase3 guard (rootless-podman aardvark-dns bind failure); root-caused live (§11.4.102), fixed the LE boot-timeout→SKIP + three latent set -e/pipefail/grep -c suite-abort bugs → the suite now **COMPLETES + honestly reports (74/60/11/3)** instead of masking failures behind a hang. **! HOST PODMAN BROKEN (env, operator-actionable, NOT a code defect):** aardvark-dns cannot bind netavark gateway :53 host-wide → proxy containers show "Up (healthy)" but crun says NOT running, :53128/:51080 don't serve, ./start fails identically; a host-level podman/netavark reset is needed but was NOT done autonomously (§11.4.101/§11.4.174 — shared host w/ operator lava-\*/deploy\_caddy/wg0-mullvad). The 3 suite FAILs are ALL this one condition. **3 subagents in flight:** control-plane coverage r3, bidirectional both-way protocol assertions, §11.4.169 concurrency+load. **(Rev-10 prior:) §11.4.126 autonomous loop + VPN-LAN service-access feature (Phases 0/2/3/4/8/9 landed; 5/6/7/10/11 in flight).** **Rev-10 wave (12 commits this session, all FF to origin, HEAD d218977):** 8140d40 S1 security-ACL SKIP→GREEN (authoritative access.log TCP\_DENIED/HIER\_NONE, standing suite 69/63/6/0, §11.4.169 matrix 10 PASS/1 SKIP); fe9d9a9 VPN-LAN comprehensive phased plan (12 phases, env-var

bridge §11.4.28, SSRF-reconciled §11.4.120, 5 cited deep-research streams §11.4.150); 12faf12 **Phase 8 Miracast §11.4.112 structurally-impossible verdict** (cited Wi-Fi-Alliance, Cast alternative); 628d255 cmd/healthd coverage 59.5→70.3% (sample 0→100%, -race clean, conductor-re-verified); d781002 **Phase 0 env-var sword bridge scaffold** (.env.example + tests/lib/sword\_bridge.sh + scripts/sword\_doctor.sh + companion doc; 3 verdicts UP/SKIP/MISCONFIGURED reproduced); a5e5616 **Phase 9 operator bridge-setup guide**; 5c28f56 **wired test\_vpn\_lan\_bridge into the standing suite** (bridge-down honest SKIP + §11.4.115 teeth UP-stub⇒UP, suite GREEN **71/64/7/0**); 182e80a **Phase 2/3 SMB/NFS + FTP/SFTP/WebDAV tests** (sha256 round-trip, wrong-answer⇒FAIL, bridge-down SKIP PASS\_lines=0); 2f31460 **Phase 4 email + open-relay guard** (external-RCPT-accepted⇒FAIL, creds via stdin never argv); d218977 **VPN-LAN integration Status + Status\_Summary** (§11.4.45/§11.4.56). **Anti-bluff pattern proven:** every protocol test sources tests/lib/sword\_bridge.sh, calls bridge\_require FIRST, honest-SKIPs (exit 0, zero ^PASS:) when the bridge is down — never a fake PASS; ab\_pass\_with\_evidence refuses a PASS without a non-empty captured artefact. **3 subagents in flight (§11.4.70/§11.4.103):** Phase 6/7 Chromecast+ADB tests, Phase 11 Challenge + HelixQA vpn\_lan.yaml bank, Phase 5 discovery-reflector design+test. **Data-plane health FACT (§11.4.7):** base proxy 204 on a real endpoint via :53128; the fake-domain cache.example 503/000 is NOT a regression. Host 43%. **(prior) §11.4.126 autonomous hardening loop** (operator: "keep hardening, don't tag yet"). **P10 VPN fail-closed = GREEN:** dynamic stack booted, tunnel DOWN → branded 503 ERR\_TUNNEL\_DOWN ×3, leak\_seen=0, Squid PID unchanged, deterministic ×3 + RED-polarity; egress-half operator-gated SKIP (§11.4.21). **2 security fixes landed + VERIFIED DEPLOYED LIVE** (§11.4.108 runtime-signature): Squid header/version-hygiene (via off+forwarded\_for delete+version-suppression — Via gone) 4f983ee; Dante SOCKS5 SSRF (block link-local/loopback/RFC1918 + command:connect) 4626f05. **Rev-7 hardening wave (5 commits):** 790c191 **S4 SSRF guard hardened** — the SOCKS-block verdict now requires dante's authoritative block(N) log line (§11.4.69), NOT elapsed-time (an independent §11.4.142 review found the timing-only discriminator bluff-capable on fast-refuse hosts); **security guard now WIRED into the standing suite** (run-tests.sh test\_security\_guards, §11.4.135, GREEN+RED-polarity, set-e-safe) — iterate-to-GO (§11.4.134); 487c918 **cache challenge authoritative** — reads Squid's own access.log via podman exec → real TCP\_MEM\_HIT (§11.4.69), no more SKIP-fallback (Squid caching proven genuine); 3755702 **control-plane unit coverage** store 64.5→98.2% / vpn 78.6→98.1% / api 69.6→80.3% / healthd 61.3→67.6% (real error/fail-closed branches, race-clean, go.sum unchanged); 86126ac README §11.4.57 doc-link section; ad720ec this file → Rev 6. **2 items TRACKED-for-operator** (connectivity-risk §11.4.101): Squid dns\_nameservers DNS-leak

(static mode); Dante client-side open-relay. **§11.4.169 matrix = 9 PASS / 2 honest SKIP** (docs/design/hardening/Status.md Rev 3). **LE issuance + renewal BOTH PROVEN** — autonomous scope COMPLETE; Phase 4/6 OPERATOR-BLOCKED (§11.4.10). **Rev-8 increment (2 commits, subagent-driven §11.4.70 + conductor-reviewed pre-commit §11.4.142):** 017482a closed the §11.4.18 companion-doc gap (16 of 61 scripts → docs/scripts/<name>.md + synced HTML+PDF, all 16 PDFs §11.4.168 leak-clean, library function-lists cross-checked against real source §11.4.6); 0e987f1 raised control-plane internal/api unit coverage 80.3→95.8% (real error/fail-closed/500/mTLS-bootstrap branches, race-clean, go.mod/go.sum unchanged). HEAD 0e987f1 (== main == github/origin/upstream). **Branch:** feature/vpn-aware-dynamic-routing **Spec:** docs/superpowers/specs/2026-06-30-vpn-aware-proxy-extension-design.md (Rev 4) **Plan:** docs/superpowers/plans/2026-06-30-vpn-aware-proxy-extension-plan.md (Rev 1) **Authority:** Inherits the Helix Constitution submodule (constitution/Constitution.md) per §11.4.35.

§12.10 live-state resume file. Read this first, then git fetch --all --prune and re-read git log --oneline main..HEAD. Any agent must be able to resume exactly where the last session left off from this single file.

---

## 1. Current PHASE

**§11.4.126 autonomous hardening loop** — construction (P0-P10 fail-closed) is landed; the loop is now closing hardening gaps under all §11.4.169 test types with real captured evidence **AND has opened a new feature workstream: VPN-LAN service access** (operator mandate — reach/use all mainstream services on the far side of the sword\_toolkit VPN). The authoritative phased tracker is docs/design/vpn\_lan\_access/PLAN.md (§11.4.172, 12 phases). Phase 0 (env-var bridge scaffold + sword-doctor) is in flight via a subagent; Phase 8 (Miracast §11.4.112 verdict) + a control-plane coverage push run in parallel. Operator decision in force: **"keep hardening, don't tag yet."** The Go control-plane (stores, health-publisher, acl-helper, config-compiler, P5b breaker/failover, P6 control-API/SSE/metrics/PAC/mTLS) builds + vets + gofmt-clean and is proven unit / integration / config-parse; the dynamic compose profile boots live and the **fail-closed data-plane proof is GREEN**. Base proxy UP on :53128 (204); host ~43%; 4 helixproxy\_\* Podman secrets present (enable dynamic re-boot).

**P10 VPN fail-closed — GREEN (§11.4.68/.115/.108):** tests/dynamic/vpn\_failclosed\_test.sh proven live — booted the dynamic stack, forced the tunnel DOWN via Redis vpn:status, tunnel-DOWN ⇒ branded 503 ERR\_TUNNEL\_DOWN ×3 (real 3132-byte page), leak\_seen=0, Squid PID unchanged; deterministic ×3 (§11.4.50) +

RED polarity guard FAILs a fabricated 200 leak (§11.4.115). Real-VPN-egress half is operator-gated SKIP (gluetun WireGuard creds §11.4.21). Evidence qa-results/dynamic/vpn\_failclosed/20260701T130115Z/. Re-runnable boot recipe: 4 external Podman secrets from tests/observability/gen\_test\_mtls.sh → ./start --dynamic (backgrounded) → poll proxy-squid Up + compiler renders dynamic-routing.squid →  
 HELIX\_DYNAMIC\_STACK=1 GOMAXPROCS=2 nice -n 19 ionice -c 3 bash tests/dynamic/vpn\_failclosed\_test.sh; restore base after: ./stop && ./start.

## **2 real security fixes (config-security review, docs/design/security/Status.md Rev 2) — RED→GREEN + sink-side evidence + standing guards:**

- Squid header/version-hygiene (4f983ee): via off + forwarded\_for delete + httpd\_suppress\_version\_string on + visible\_hostname helix-proxy in squid.conf + squid.dynamic.conf. The Via: 1.1 proxy-squid (squid/6.13) leak was **CONFIRMED** live (RED) → GONE (GREEN). Guard: proxy\_acl\_security.sh S3. **Root-cause note:** single-file :ro bind mounts pin the inode → config edits need a container **recreate** (./stop && ./start), NOT squid -k reconfigure (re-reads the stale inode).
- Dante SOCKS5 SSRF (4626f05): command: connect + socks block for 127/8, 169.254/16, 10/8, 172.16/12, 192.168/16 in sockd.conf. 5 internal targets refused fast (code 000 ~0.01s, dante-log block(N)), external control 204, no public-egress regression. Guard: proxy\_acl\_security.sh S4.

**2 items TRACKED-for-operator (§11.4.101 connectivity-risk, NOT autonomously fixed):** (1) Squid dns\_nameservers 8.8.8.8 bypasses the DoT dnsproxy → DNS leak in **static** mode (dynamic mitigated by never\_direct); re-point = risk. (2) Dante socksmethod none + client pass from:0.0.0.0/0 open-relay if :51080 escapes the bridge; client-CIDR restriction = risk. O(1) table in docs/design/security/Status.md.

**§11.4.169 hardening matrix (docs/design/hardening/Status.md Rev 3) — 9 PASS / 2 honest SKIP:** PASS = stress+chaos, DDoS(300/300), concurrency(40, crosstalk=0), memory(ratio 1.0017), **P10 fail-closed, race/deadlock (0 DATA RACE, 11 pkgs), benchmark (200/200, p50=86ms/p95=88ms/p99=91ms, 10.84 req/s)** + unit/integration. SKIP (honest §11.4.3) = security ACL live-deny (no autonomous deny topology) + P10 egress-half (gluetun creds).

**Remaining actionable (non-operator-gated) is thinning.** Operator-gated: 2 TRACKED security items (connectivity risk), LE Phase 4/6, P10 real-egress (gluetun creds), HelixQA vendoring (6 un-vendored siblings), release tag.

## 2. Landed commits (newest first)

**main FF-tracks HEAD (§11.4.113 FF-only), so main..HEAD is empty — HEAD == main == github/origin/upstream == 0e987f**

1. The full feature-branch history since the original branch point is below; the historical table (numbered 1–28) is retained for the earlier construction wave.

### Latest hardening wave (newest first):

| Commit  | Lane          | Summary                                                                                                                                                                                              |
|---------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0e987f1 | control-plane | internal/api unit coverage 80.3→95.8% (real 404/400/500/502/fail-closed-TLS/metrics-suppression branches, race-clean, go.mod unchanged) — subagent-driven (§11.4.70), conductor-reviewed (§11.4.142) |
| 017482a | docs          | 16 §11.4.18 companion docs (previously-undocumented test/challenge/lib scripts) + synced HTML+PDF, all §11.4.168 leak-clean, library fn-lists source-verified (§11.4.6)                              |
| 790c191 | security      | S4 SSRF guard → authoritative dante block(N) log-line discriminator (§11.4.69) + security guard wired into standing suite (§11.4.135) — independent-review-driven (§11.4.142/.134)                   |
| 487c918 | challenge     | proxy cache challenge → authoritative TCP_*HIT via container access.log (§11.4.69), no more SKIP-fallback (Squid caching proven genuine)                                                             |
| 3755702 | control-plane | unit coverage store 64.5→98.2% / vpn 78.6→98.1% / api 69.6→80.3% / healthd 61.3→67.6% (real error/fail-closed branches, race-clean)                                                                  |
| 86126ac | docs          | README §11.4.57 Tracked-Items doc-link section (9 Status docs)                                                                                                                                       |
| ad720ec | docs          | CONTINUATION → Rev 6 (§12.10 sync: P10 GREEN + security fixes + §11.4.169 matrix)                                                                                                                    |
| 4626f05 | security      | Dante SOCKS5 SSRF hardening — block internal/link-local/loopback egress + command:connect (RED→GREEN + S4 guard)                                                                                     |
| 4d0a7ed | control-plane | drop dead nil-check in TestPostgresSatisfiesQueries (golangci SA4023)                                                                                                                                |
| 916e72b | helixqa       |                                                                                                                                                                                                      |

| Commit  | Lane        | Summary                                                                                                              |
|---------|-------------|----------------------------------------------------------------------------------------------------------------------|
|         |             | unlock recipe for the proxy test bank (6 un-vendored own-org siblings)                                               |
| 8833ccc | letsencrypt | cert-analyzer edge-case coverage 37→55 (validity boundaries, malformed PEM, empty/IP/mixed SAN, double-wildcard)     |
| 4f983ee | security    | Squid header/version-hygiene hardening (via off + forwarded_for delete + version suppression) — RED→GREEN + S3 guard |
| 1caaf51 | hardening   | control-plane unit(100-61%)+Go-benchmarks+audit-atomicity-verified + Challenges 2/3 + §11.4.169 matrix sync          |
| 567c9e1 | hardening   | P10 VPN fail-closed GREEN + race(0)/benchmark(p50=86ms) + §11.4.169 Status matrix                                    |

**Earlier construction wave (historical, numbered from the original branch point):**

| #  | Commit  | Phase       | Summary                                                                                                             |
|----|---------|-------------|---------------------------------------------------------------------------------------------------------------------|
| 28 | 8d95f8a | P6          | real bidirectional metric-name drift guard (§1.1-mutation-proven) + concurrency consistency test; WARNING-3/4/5     |
| 27 | 2bc03de | BUGFIX-0006 | revive + de-bluff comprehensive-test.sh ((( )) abort = 100% dead) + real B2/B3/B8 evidence; surfaced regression #50 |
| 26 | 4394643 | BUGFIX-0005 | final-verify.sh + verify-proxy.sh no longer green a NO-VPN config (false-VPN-routing §15) + set -e abort            |
| 25 | cd11494 | BUGFIX-0004 | run-tests.sh no longer FAILs a healthy proxy — §11.4.3 topology-aware ports + 3-state SKIP                          |
| 24 | 6a8f886 | P11         | refresh CONTINUATION to live state (Rev 2) — 23 commits, P5b/P6/BUGFIX-0002/0003 landed, P8 in flight               |
| 23 | 61b4215 | chore       | gofmt-format 6 pre-existing files (formatters-clean mandate; semantics-null verified)                               |
| 22 | 62b22fe | P6          | control-API server (REST/SSE/metrics/PAC, fail-closed mTLS) +                                                       |

| #  | Commit  | Phase       | Summary                                                                                            |
|----|---------|-------------|----------------------------------------------------------------------------------------------------|
|    |         |             | coherent operator-wiring contract                                                                  |
| 21 | 1045dfd | BUGFIX-0003 | test_result must return 0 — suite no longer aborts mid-run under set -e                            |
| 20 | c6f2935 | P9          | §11.4.18 operator-guide companions for the 16 tests/dynamic scripts                                |
| 19 | b5573a9 | BUGFIX-0002 | squid log-dir writable under rootless Podman (proxy crash-loop) — existing features now serve live |
| 18 | 0aca034 | P9          | anti-bluff dynamic-routing test/analyzer harness (tests/dynamic)                                   |
| 17 | e6e93ec | P10-prep    | dynamic compose profile + control-plane/squid Containerfiles + orchestrator wiring                 |
| 16 | 6bdeef9 | P5b         | circuit-breaker + tier-failover (internal/breaker, gobreaker/v2)                                   |
| 15 | 7d0d128 | P11         | CONTINUATION (§12.10) + spec §9 reconcile + §11.4.65 HTML/PDF export backfill                      |
| 14 | 1833c8f | P7.3        | per-user Squid auth + rootless Podman-secret loader + kill-switch design (no secrets)              |
| 13 | 603e039 | P5a         | acl-helper — Squid external_acl OK/ERR from Redis, fail-closed (stdlib)                            |
| 12 | e6e336f | P7.1        | per-tunnel DoH/DoT (dnsproxy) config plan + DNS-leak test design                                   |
| 11 | 04526dd | P4          | config-compiler — render Squid/Dante/PAC from PG + seed route keys (parse-verified)                |
| 10 | 833fb9e | P7.2        | Prometheus scrape + Grafana dashboard config plan (promtool-validated)                             |
| 9  | 11106a4 | P3          | vpn-health-publisher (cmd/healthd + internal/vpn) — data-plane health, fail-closed, TDD            |
| 8  | b66d172 | P4          | Squid 6.13 + Dante dynamic-mode templates (additive, parse-verified) + spec reconcile              |
| 7  | fbfe9ed | P1          |                                                                                                    |

| # | Commit  | Phase | Summary                                                                                                                                            |
|---|---------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | e19e0ed | P2    | docs(spec): mark §20 gaps G1-G4<br>RESOLVED with spike decisions<br>store (pgx) + redis (go-redis)<br>clients — fail-closed, TDD, real<br>PG/Redis |
| 5 | 6409cb9 | P1    | docs(research): resolve spec §20<br>gaps G1-G4 with captured-<br>evidence spikes                                                                   |
| 4 | 6802798 | P1/E  | docs(audit): §11.4.138 forensic<br>bluff-audit of 4 existing test<br>scripts (8 bluffs)                                                            |
| 3 | 9ac1b4a | P0    | docs(dynamic-routing):<br>DYNAMIC_ROUTING.md + 2<br>mermaid diagrams                                                                               |
| 2 | 6251007 | P0    | chore(submodules): incorporate<br>containers, helix_qa, challenges,<br>docs_chain (SSH, no-force)                                                  |
| 1 | 5f917a7 | P0    | P0 scaffold — data model,<br>evidence harness, Go skeleton,<br>governance carriers                                                                 |

### 3. PROVEN-NOW vs OWED-TO-P10 (honest §11.4.6)

#### PROVEN-NOW (control-plane / config-plane / spike facts — captured)

- **Existing proxy serves LIVE (BUGFIX-0002)** — after the rootless-Podman log-dir fix, the booted --no-vpn stack proves all 3 existing features: HTTP forward proxy 200 + Via: 1.1 proxy-squid, Dante SOCKS5 200, squid cache TCP\_MEM\_HIT (no origin contact). Guard: tests/regression/log\_dir\_writable\_test.sh (§11.4.115 polarity, §1.1 mutation byte-identical md5 0128a96b6d467c2da5b7cef8a808e563). Evidence: qa-results/regression/bugfix38/.
- **P5b breaker/failover** (internal/breaker, gobreaker/v2) — per-target circuit breaker + tunnel tier-failover, TDD.
- **P6 control-API** (cmd/api + internal/api + internal/pac) — REST CRUD + SSE + Prometheus /metrics + PAC, **fail-closed mTLS** (RequireAndVerifyClientCert), coherent operator-wiring contract (CONTROL\_API\_TLS\_CERT/\_KEY/\_TLS\_CLIENT\_CA, :58080); builds + vets clean, §1.1 mutation md5 67125c7a1ab9b00c98fb164f765b04af.
- **Spec §20 gaps G1-G4 resolved** with transient-spike captured evidence (docs/research/mvp/findings/F\_spikes\_G1-G4.md, run-id



qa-results/spikes/20260630\_205029\_glg4/): G2 ubuntu/  
 squid:latest = Squid **6.13** (not v8), §8 directive set squid -k  
 parse exit 0; G4 gluetun **v3.40 (=v3.40.4)** control-API :8000  
 answers 200, issue #3060 confirmed; G1 kernel-WG interface  
**creatable rootless** with --cap-add NET\_ADMIN; G3 Dante  
**SIGHUP preserves an active SOCKS session** (20/20  
 chunks, curl exit 0, /proc/net/tcp ESTABLISHED proof).

- **P2 stores** (pgx + go-redis) — fail-closed, TDD, exercised against **real PG / Redis**.
- **P3 vpn-health-publisher** (cmd/healthd + internal/vpn) — data-plane health poll → Redis state, fail-closed, TDD.
- **P4 config-compiler + templates** — Squid 6.13 (%>ha{Host}) + Dante (concatenation, no include) render from PG; **squid -k parse exit 0**; PAC + route-key seeding parse-verified.
- **P5a acl-helper** — Squid external\_acl OK/ERR from Redis, **fail-closed**, stdlib-only.
- **P7.2 observability** config plan — **promtool-validated** Prometheus scrape + Grafana dashboard.
- **P7.1 DNS / P7.3 security** — config plans only (DoH/DoT per-tunnel; per-user auth + Podman-secret loader + in-netns kill-switch). Design + parse layer.
- **Existing-test bluff audit** (Stream E) — 8 bluffs across 4 scripts catalogued (§11.4.138), guards owed to P8.

## **CAPTURED-AT-P10 (fail-closed data-plane proof — the dynamic stack HAS booted)**

The dynamic compose profile (postgres + redis + control-plane + squid+helper + dante) now **boots live**; the fail-closed half of the usability proof is captured:

- graceful\_503 — **PROVEN**: tunnel DOWN (Redis vpn:status) → branded 503 ERR\_TUNNEL\_DOWN ×3 (3132-byte page) with **Squid PID unchanged**; deterministic ×3 + RED-polarity guard. Evidence qa-results/dynamic/vpn\_failclosed/20260701T130115Z/.
- no\_leak (tunnel-down case) — **PROVEN**: leak\_seen=0 during the DOWN window (no target reached while the tunnel is down).

## **OWED-TO-P10 (real-VPN-egress half — operator-gated on gluetun WG creds)**

Requires real gluetun WireGuard credentials (§11.4.21); still **unproven live**:

- vpn\_real\_egress — egress IP via proxy  
 == tunnel exit && != host IP + **wg transfer Δ** (200 OK is not routing).
- no\_leak / killswitch (up→drop case) — drop a *real* tunnel → **zero** target packets on the real uplink (tcpdump) + DNS only via the intended resolver.

- per-user **407 auth challenge** live; **secret injection leak-free** at runtime.
- **G1 residual** — full rootless kernel-WG *operation* (handshake + routing + throughput), only interface *creation* was spiked (§20 G1).
- **G3 residual / P9** — concurrent / repeated SIGHUP + **route-change-mid-session** SOCKS path behaviour (§20 G3).
- circuit-breaker open → failover to next up tier — **landed** (6bdeef9, P5b); live-under-load failover proof still owed.

## 4. Remaining phases

| Phase      | Scope                                                                                                | State                                                                                                  |
|------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>P5b</b> | per-target circuit breaker + tunnel tier-failover (sony/gobreaker/v2)                                | ✓ landed 6bdeef9                                                                                       |
| <b>P6</b>  | control-API + SSE + metrics + PAC + fail-closed mTLS                                                 | ✓ landed 62b22fe (admin-UI templ/htmx + §11.4.170 host-rendered pixel proof = P6.2, deferred)          |
| <b>P8</b>  | fix existing-test bluffs → §11.4.3 topology dispatch / honest SKIP / §11.4.161 + §11.4.135 guards    | ✓ landed (cd11494/4394643/2bc03de/8d95f8a)                                                             |
| <b>P9</b>  | full test matrix + Challenges + HelixQA (all §11.4.169 types; G3 route-change-mid-session live test) | §11.4.169 matrix GREEN (9 PASS/2 SKIP); Challenges 2/3; HelixQA vendoring operator-gated (6 siblings)  |
| <b>P10</b> | <b>live dynamic-mode boot + captured data-plane evidence = the usability proof</b>                   | ✓ <b>fail-closed half GREEN</b> (567c9e1); real-egress half operator-gated (gluetun WG creds §11.4.21) |
| <b>P11</b> | docs sync + HTML/PDF (+DOCX where mandated) exports (this CONTINUATION + .remember are part of it)   | ongoing (this Rev 6 sync)                                                                              |
| <b>P12</b> | whole-branch review (iterate-to-GO) + full retest + merge to main no-force + prefixed release tag    | operator-gated — "keep hardening, don't tag yet"                                                       |

## 5. Binding constraints (non-negotiable)

- **Anti-bluff §11.4** — every PASS carries positive captured **data-plane** evidence; control-plane/config-parse green is necessary, never sufficient; the end-user-usability bar is met only at P10.
- **No force-push §11.4.113** — merge onto latest main, fast-forward only; force-push is forbidden with no exception.
- **Rootless Podman §11.4.161** — all containers rootless; no Docker-rootful, no sudo, no root escalation; orchestrate via the containers submodule (§11.4.76), build on the remote host (§11.4.173).
- **Secrets-as-names-only §11.4.10** — VPN creds / proxy-auth / mTLS keys via Podman secrets / file refs; **never** plaintext in git; `.env.example` documents refs only.
- **Operator-safe §11.4.174** — do **NOT** touch the operator's pre-existing resources: the host `wg0-mullvad` (UP kernel-WG) interface and any `lava-*` containers (e.g. `lava-postgres-thinker`) are off-limits; verify process/resource ownership before acting; block-don't-break on shared-host contention.
- **Host safety §12** —  $\leq 60\%$  memory (§12.6); no host power-state commands (CONST-033); pull images sequentially; `--rm` diagnostics; `df` first.

## 6. Resume now (next actionable)

1. `git fetch --all --prune` on `feature/vpn-aware-dynamic-routing`; confirm HEAD `0e987f1` (`== main ==` all 3 remotes `github/origin/upstream`; integrate any newer foreign commit per §11.4.71, no force §11.4.113). The single canonical moment-valid resume file is `.remember/remember.md` (§11.4.131) — read it first.
2. **Continue the §11.4.126 autonomous hardening loop** (operator: "keep hardening, don't tag yet"). Base proxy UP :53128 (204); ~43% host; 4 `helixproxy_*` Podman secrets present. Keep dispatching 3-4 parallel non-data-plane subagents on remaining actionable items (§11.4.103); the data plane / :53128 has a single owner (§11.4.119) — coordinate before any boot.
3. **P10 fail-closed — GREEN + guarded (567c9e1)**. Real-VPN-egress half remains operator-gated on `gluetun WireGuard` creds (§11.4.21/.66). To re-run fail-closed: 4 secrets from `tests/observability/gen_test_mtls.sh` → `./start --dynamic` (backgrounded) → `HELIX_DYNAMIC_STACK=1 ... bash tests/dynamic/vpn_failclosed_test.sh` → `restore base ./stop && ./start`.
4. **LE — issuance + renewal BOTH PROVEN (autonomous scope COMPLETE)**. Phase 3 hermetic DNS-01 issuance + Phase 5 zero-downtime renewal/rotation are cert-analyzer-verified, re-runnable, and guarded (`tests/letsencrypt/phase3_issuance_guard.sh` + `phase5_rotation_guard.sh`, wired in `run-tests.sh`); custom Caddy image via `deploy/letsencrypt/`

build.sh; cert-analyzer self-test 37→55 (8833ccc). Phase 4 (LE-staging token §11.4.10)

- Phase 6 (prod domain) OPERATOR-BLOCKED. Docs docs/design/letsencrypt/Status.md.

5. **Operator-gated queue (surface, don't autonomously break):** 2 TRACKED security items (Squid dns\_nameservers DNS-leak, Dante client-side open-relay — both connectivity-risk §11.4.101); LE Phase 4/6; P10 real-egress (gluetun creds); HelixQA vendoring (6 un-vendored siblings, docs/helixqa/UNBLOCK.md); the release tag helix\_proxy-0.1.0-dev-0.0.2 (operator said don't tag yet).
6. Every change: TDD reproduce-first (§11.4.43/§11.4.115), all warranted test types (§11.4.169), paired §1.1 mutation, independent review → iterate-to-GO (§11.4.142/§11.4.125/§11.4.134), docs in sync (§11.4.60/§11.4.65/§11.4.106), operator resources untouched (§11.4.174: wg0-mullvad, lava-\*, whoami:58080).